Deep Learning-Aided DoA Estimation with Uncertainty Extraction

Raz Zohar, Shai Ginzach, and Nir Shlezinger

***Abstract*—Direction of arrival (DoA) estimation often requires not only accurate predictions but also reliable measures of uncertainty. While deep learning methods enhance robustness in challenging scenarios, they rarely provide such quantification. In this work, we show that DoA estimators based on the SubspaceNet methodology, which augments subspace methods with deep learning, naturally enable uncertainty extraction without requiring any architectural modifications. By leveraging the preserved subspace structure, we adapt classical asymptotic error analysis of ESPRIT to yield principled uncertainty estimates alongside DoA recovery. We propose an uncertainty-augmented SubspaceNet framework and demonstrate through extensive simulations that it provides both accurate DoA estimation and faithful uncertainty measures, outperforming classical subspace methods in adverse scenarios.**

***Index Terms*—Uncertainty extraction, DoA estimation.**

I. Introduction

Direction of arrival (DoA) estimation is a fundamental task in array signal processing, underlying a broad range of applications in radar, sonar, wireless communications, and acoustic source localization [1]. In many scenarios, it is not sufficient to merely provide a point estimate of the impinging signal directions; rather, one is often also interested in quantifying the reliability of these estimates. Such uncertainty information plays a critical role in enabling downstream decision-making, sensor management, and tracking algorithms to account for estimation confidence.

A leading family of algorithms for DoA estimation is based on subspace methods, such as MUSIC [2] and ESPRIT [3]. These techniques exploit the orthogonality between the signal and noise subspaces of the input covariance to achieve angular resolution that is not limited by the array geometry [4]. In addition to their interpretable spectral representation, subspace methods can provide measures of uncertainty, in the sense of predicting the error variance associated with each estimated direction [5], [6]. However, the accuracy and reliability of these methods hinge on key assumptions, including the availability of sufficiently many independent snapshots, non-coherent sources, and precisely calibrated arrays. When these conditions are violated, as often occurs in practical settings, the performance of conventional subspace methods can degrade sharply.

The emergence of deep learning has lead to the development of deep neural network (DNN)-based DoA estimation methods, that can learn from data to operate in challenging scenarios where classical subspace methods struggle [7]. Black-box DNNs based on multi-layer perceptrons (MLPs) [8]–[10], convolutional neural networks (CNNs) [11]–[13], and attention models [14], [15], were proposed to map array measurements or covariance features into DoAs. An alternative approach follows model-based deep learning methodologies [16], using DNNs into established array processing algorithms rather than replacing them. DNN augmentation were proposed to provide MUSIC spectrum [17], [18], as well as to produce covariance features, either learning from ground truth clear covariance as in [19]–[21], or by seeking a surrogate covariance for downstream DoA recovery as in SubspaceNet [22] and related algorithms [23]–[26]. The latter approach allows subspace methods to operate in settings not holding standard assumptions, such as coherency and calibration. Still, a core limitation is that DNNs typically do not provide measures of uncertainty alongside their predictions. Incorporating uncertainty quantification into DNNs generally requires substantial architectural modifications, such as those introduced in Bayesian deep learning [27] or conformal prediction frameworks [28].

In this work, we identify a notable advantage of model-based deep learning approaches for DoA estimation based on Subspace, in their ability to not only recover the source directions but also to extract reliable measures of uncertainty. This stems from the fact that such architectures preserve the core processing steps of conventional subspace methods, enabling one to directly apply established uncertainty extraction techniques [5], [6] without requiring any modification to the deep learning model itself. We systematically demonstrate that our methodology yields accurate DoA estimates along with faithful uncertainty predictions, both in terms of empirical error and predicted error magnitude. Furthermore, our results show that this capability is retained not only in favorable conditions where classical algorithms succeed, but also in challenging scenarios where subspace methods struggle.

The rest of this paper is organized as follows: Section II presents the system model and reviews some preliminaries. We derive the uncertainty extraction method in Section III and evaluate it in Section IV. Section V provide concluding remarks.

II. System Model and Preliminaries

A. Problem Formulation

Signal Model: We consider a uniform linear array (ULA) equipped with N elements, spaced at half-wavelength intervals, operating in the far-field. The array receives M static, stationary, narrowband source signals over T temporal snapshots. Let $\boldsymbol{X} \triangleq [\boldsymbol{x}(1), \ldots, \boldsymbol{x}(T)] \in \mathbb{C}^{N \times T}$ denote the matrix of baseband samples measured at the array elements, where $\boldsymbol{x}(t) \in \mathbb{C}^N$ is the array snapshot at time index t. The source signals are collected in the matrix $\boldsymbol{S} \triangleq [\boldsymbol{s}(1), \ldots, \boldsymbol{s}(T)] \in \mathbb{C}^{M \times T}$, where $\boldsymbol{s}(t) \in \mathbb{C}^M$ is the vector of source symbols at time t, impinging

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on the array from DoAs $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_M]^{\mathsf{T}}$. Under the standard narrowband and far-field assumptions, it holds that

$$\boldsymbol{X} = \boldsymbol{A}(\boldsymbol{\theta})\boldsymbol{S} + \boldsymbol{V}, \tag{1}$$

where $\boldsymbol{V}$ is the noise matrix with i.i.d. zero-mean entries of variance σ_v^2, and $\boldsymbol{A}(\boldsymbol{\theta}) \triangleq \big[\boldsymbol{a}(\theta_1), \ldots, \boldsymbol{a}(\theta_M)\big] \in \mathbb{C}^{N\times M}$ is the array steering matrix, with $\boldsymbol{a}(\theta) \in \mathbb{C}^N$ being the steering vector corresponding to a signal impinging from direction θ, defined as

$$\boldsymbol{a}(\theta) \triangleq \big[1,\, e^{-j\pi\sin(\theta)},\, \ldots,\, e^{-j\pi(N-1)\sin(\theta)}\big]^{\mathsf{T}}. \tag{2}$$

DoA Estimation with Uncertainty: Our goal is to design an algorithm that maps the array measurements $\boldsymbol{X}$ into both (i) An estimate $\hat{\boldsymbol{\theta}}$ of the true DoAs $\boldsymbol{\theta}$, evaluated in the mean-squared error (MSE) sense , i.e., $\mathbb{E}\big[\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2\big]$; and (ii) an estimate $\hat{\boldsymbol{\sigma}}$ of the accuracy (uncertainty) associated with each recovered DoA, whose fidelity is evaluated via the MSE between the predicted estimation error and the empirical estimation error magnitude, i.e., $\mathbb{E}\big[\|\hat{\boldsymbol{\sigma}} - |\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}|\|^2\big]$, where $|\cdot|$ is taken element-wise.

We target scenarios in which the algorithm must operate reliably under conditions that are known to be challenging, including:

C1 *Coherent sources*, whose covariance is rank-deficient.

C2 *Array miscalibration* which deviate from the assumed (2).

C3 *Limited snapshots*, i.e., a small number of temporal observations (T) or low signal-to-noise ratio (SNR) (large σ_v^2).

To address these challenges, we assume access, during the design stage, to a labeled dataset containing array measurements and their corresponding ground-truth DoAs, denoted

$$\mathcal{D} = \big\{\boldsymbol{X}^{(i)}, \boldsymbol{\theta}^{(i)}\big\}_{i=1}^{|\mathcal{D}|}. \tag{3}$$

B. Preliminaries

Subspace Methods: A widely used class of DoA estimators exploits the decomposition of the spatial covariance matrix into signal and noise subspaces. Let $\boldsymbol{R}_S$ denote the covariance of the source signals $\boldsymbol{s}(t)$. From (1), the array covariance is

$$\boldsymbol{R}_X = \mathbb{E}\big[\boldsymbol{x}(t)\boldsymbol{x}^{\mathsf{H}}(t)\big] = \boldsymbol{A}(\boldsymbol{\theta})\,\boldsymbol{R}_S\,\boldsymbol{A}^{\mathsf{H}}(\boldsymbol{\theta}) + \sigma_v^2\boldsymbol{I}_N, \tag{4}$$

where $\boldsymbol{R}_S \triangleq \mathbb{E}[\boldsymbol{s}(t)\boldsymbol{s}^{\mathsf{H}}(t)]$. When $\boldsymbol{R}_S$ is full rank (not holding *C1*), then the eigenvectors matrix of $\boldsymbol{R}_X$ can be partitioned into the *signal subspace* $\boldsymbol{U}_S$, spanned by the steering vectors, and the *noise subspace* $\boldsymbol{U}_N$, orthogonal to them. This orthogonality implies that $\|\boldsymbol{U}_N^{\mathsf{H}}\,\boldsymbol{a}(\theta_i)\|^2 = 0$, forming the basis for subspace methods, which use the empirical estimate $\hat{\boldsymbol{R}}_X = \frac{1}{T}\boldsymbol{X}\boldsymbol{X}^{\mathsf{H}}$ and the steering vectors form in (2) (which are inaccurate under *C3* and *C2*, respectively) to obtain the DoAs from this relationship.

ESPRIT: A representative subspace method is ESPRIT [3], which exploits the rotational invariance property of the signal subspace. For a ULA with N sensors, the steering matrix can be viewed as two maximally overlapping subarrays with steering matrices $\mathbf{A}_1(\boldsymbol{\theta})$ and $\mathbf{A}_2(\boldsymbol{\theta})$ related via $\mathbf{A}_2(\boldsymbol{\theta}) = \mathbf{A}_1(\boldsymbol{\theta})\,\mathbf{E}$, where $\boldsymbol{E} = \mathrm{diag}\big(e^{-j\pi\sin(\theta_1)}, \ldots, e^{-j\pi\sin(\theta_M)}\big)$. Define the subarray selection matrices $\boldsymbol{J}_1, \boldsymbol{J}_2 \in \{0,1\}^{(N-k)\times N}$, where k is the displacement between the two overlapping subarrays (for maximal overlap, $k=1$). The truncated signal subspaces for the first and second subarrays are $\boldsymbol{E}_x = \boldsymbol{J}_1\boldsymbol{U}_S$ and $\boldsymbol{E}_y = \boldsymbol{J}_2\boldsymbol{U}_S$.

The rotational invariance property implies that $\boldsymbol{E}_y = \boldsymbol{E}_x\,\boldsymbol{F}$, for some nonsingular transformation matrix $\boldsymbol{F} \in \mathbb{C}^{M\times M}$. In practice, $\boldsymbol{F}$ is obtained as $\boldsymbol{F} = \boldsymbol{E}_x^{\dagger}\boldsymbol{E}_y$, where $(\cdot)^{\dagger}$ denotes the Moore–Penrose pseudoinverse. The eigenvalues of $\boldsymbol{F}$ are the diagonal entries of $\boldsymbol{E}$, and the DoAs are recovered as

$$\hat{\boldsymbol{\theta}} = \mathrm{ESPRIT}\Big(\hat{\boldsymbol{R}}_X\Big) = \sin^{-1}(\angle\,\mathrm{eig}\{\boldsymbol{F}\}/\pi)\,, \tag{5}$$

where $\mathrm{eig}\{\cdot\}$ denotes the set of eigenvalues and $\angle(\cdot)$ is the argument (phase) operator.

Deep Learning Approaches: Deep learning methods can enable DoA estimation while coping with *C1*-*C3*. The direct approach uses *end-to-end* deep learning that is invariant of subspace processing, training a DNN to directly output $\hat{\boldsymbol{\theta}}$ from the measurements $\boldsymbol{X}$ (or from features extracted from it, such as $\frac{1}{T}\boldsymbol{X}\boldsymbol{X}^{\mathsf{H}}$ [8], [11]). By letting $\boldsymbol{\psi}$ be the DNN parameters and $\mathrm{DNN}(\cdot;\boldsymbol{\psi})$ denote its mapping, the DoAs are estimated as

$$\hat{\boldsymbol{\theta}}^{\mathrm{E2E}} = \mathrm{DNN}(\boldsymbol{X};\boldsymbol{\psi}). \tag{6}$$

An alternative, *model-based* deep learning paradigm retains classical processing stages while enhancing them with learnable components [29]. Specifically, SubspaceNet [22] and its variants [25], [26], train a DNN to map the observed $\boldsymbol{X}$ into a surrogate covariance matrix $\tilde{\boldsymbol{R}}_X$, by viewing the overall processing chain including downstream subspace-based DoA estimation as a discriminative machine learning (ML) architecture [30]. When employing ESPRIT, the estimated DoAs are obtained as

$$\hat{\boldsymbol{\theta}}^{\mathrm{SSN}} = \mathrm{ESPRIT}\left(\tilde{\boldsymbol{R}}_X = \mathrm{DNN}(\boldsymbol{X};\boldsymbol{\psi})\right). \tag{7}$$

By training the DNN to minimize the DoA estimation error induced by the subspace method, the learned covariance can allow subspace-based DoA estimation under *C1*-*C3*.

III. Uncertainty Extraction

In this section we introduce our framework for simultaneously providing DoA estimates and their uncertainty under *C1*-*C3*. We note that while both methodologies (6) and (7) can learn to estimate $\boldsymbol{\theta}$ under *C1*-*C3*, they substantially differ in their ability to provide the uncertainty $\boldsymbol{\sigma}$. For end-to-end deep learning, uncertainty extraction requires altering the architecture and the training architecture, e.g., by having additional uncertainty output features, converting to Bayesian deep learning [31], or incorporating conformal prediction [28]. However, for (7), the fact that inference is carried out using ESPRIT allows seamless uncertainty extraction without altering the DNN, based on principled approximations of established asymptotic properties of ESPRIT [5].

A. ESPRIT Asymptotic Error

Our formulation is based on the rigorous characterization of the error of ESPRIT (in settings not holding *C1*-*C3*), in the asymptotic regime of large T. This is stated in the following theorem, proven in [5]:

Theorem 1: Consider a calibrated ULA with half-wavelength spacing and sufficient measurements from non-coherent far-field sources. Let $(\lambda_i, \boldsymbol{v}_i, \boldsymbol{q}_i)$ be the ith eigenvalue and right/left eigenpairs of $\boldsymbol{F}$ with $\boldsymbol{q}_i^{\mathsf{H}}\boldsymbol{v}_i = 1$, and define $\mathbf{W}_i \triangleq \mathbf{J}_1 - \lambda_i^*\,\mathbf{J}_2$,

$\mathbf{U}_i \triangleq \mathbf{J}_2 - \lambda_i \mathbf{J}_1$. Furthermore, let $(\alpha_i, \boldsymbol{\rho}_i)$ be the ith eigenvalue and eigenvector of $\boldsymbol{R}_X$, and for each $g, h \in \{1, \ldots, M\}$, define

$$\boldsymbol{\Sigma}^{(\mathrm{H})}_{gh} = \sum_{\substack{l=1\\ l\neq g}}^{N} \sum_{\substack{n=1\\ n\neq h}}^{N} \sum_{a_1,a_2=1}^{N} \sum_{b_1,b_2=1}^{N} [\boldsymbol{\rho}^*_l]_{a_1} [\boldsymbol{\rho}_g]_{a_2} [\boldsymbol{\rho}_n]_{b_1} [\boldsymbol{\rho}^*_h]_{b_2} \times \frac{\mathbb{E}\left\{[\hat{\boldsymbol{R}}_X - \boldsymbol{R}_X]_{a_1 a_2} [\hat{\boldsymbol{R}}_X - \boldsymbol{R}_X]^*_{b_1 b_2}\right\}}{(\alpha_g - \alpha_l)(\alpha_h - \alpha_n)} \boldsymbol{\rho}_l \boldsymbol{\rho}^H_n, \quad (8)$$

and

$$\boldsymbol{\Sigma}^{(\mathrm{T})}_{gh} = \sum_{\substack{l=1\\ l\neq g}}^{N} \sum_{\substack{n=1\\ n\neq h}}^{N} \sum_{a_1,a_2=1}^{N} \sum_{b_1,b_2=1}^{N} [\boldsymbol{\rho}^*_l]_{a_1} [\boldsymbol{\rho}_g]_{a_2} [\boldsymbol{\rho}_n]_{b_1} [\boldsymbol{\rho}_h]_{b_2} \times \frac{\mathbb{E}\left\{[\hat{\boldsymbol{R}}_X - \boldsymbol{R}_X]_{a_1 a_2} [\hat{\boldsymbol{R}}_X - \boldsymbol{R}_X]_{b_1 b_2}\right\}}{(\alpha_g - \alpha_l)(\alpha_h - \alpha_n)} \boldsymbol{\rho}_l \boldsymbol{\rho}^T_n. \quad (9)$$

Then, by defining $\boldsymbol{\Sigma}^{(\mathrm{H})}_i = \sum_{g=1}^{M} \sum_{h=1}^{M} [\boldsymbol{v}_i]_g [\boldsymbol{v}_i]^*_h \boldsymbol{\Sigma}^{(\mathrm{H})}_{gh}$ and $\boldsymbol{\Sigma}^{(\mathrm{T})}_i = \sum_{g=1}^{M} \sum_{h=1}^{M} [\boldsymbol{v}_i]_g [\boldsymbol{v}_i]_h \boldsymbol{\Sigma}^{(\mathrm{T})}_{gh}$ for each $i \in \{1, \ldots, M\}$, the DoAs estimated by ESPRIT (5) hold

$$\mathrm{var}(\hat{\theta}_i) = \frac{1}{2\pi^2 \cos^2 \theta_i} \Bigg(\boldsymbol{q}^{\mathsf{H}}_i \boldsymbol{E}^{\dagger}_x \boldsymbol{W}_i \boldsymbol{\Sigma}^{(\mathrm{H})}_i \boldsymbol{W}^{\mathsf{H}}_i (\boldsymbol{E}^{\dagger}_x)^{\mathsf{H}} \boldsymbol{q}_i - \mathrm{Re}\left\{ \boldsymbol{q}^{\mathsf{H}}_i \boldsymbol{E}^{\dagger}_x \boldsymbol{U}_i \boldsymbol{\Sigma}^{(\mathrm{T})}_i \boldsymbol{U}^{\mathsf{T}}_i (\boldsymbol{E}^{\dagger}_x)^{\mathsf{T}} \boldsymbol{q}_i (\lambda^*_i)^2 \right\} \Bigg). \quad (10)$$

B. Uncertainty Extraction Algorithm

Theorem 1 characterizes the variance of the DoA estimates produced by Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT). We next show how it can be approximated to provide uncertainty quantification for DNN-aided DoA estimation based on SubspaceNet (7), which provides both an estimate of the covariance $\tilde{\boldsymbol{R}}_X$ (via a DNN) and of the DoAs $\hat{\boldsymbol{\theta}}$, as well as ESPRIT's matrices $\boldsymbol{J}_1, \boldsymbol{J}_2, \boldsymbol{F}$.

Specifically, in addition to $\boldsymbol{J}_1, \boldsymbol{J}_2, \boldsymbol{F}$, Theorem 1 requires one to provide: (1) the truth DoA θ_i (to evaluate $\cos^2 \theta_i$); (2) the input covariance $\boldsymbol{R}_X$ (to compute $(\alpha_i, \boldsymbol{\rho}_i)$); and (3) the expectations

$$\mathcal{C}^{(\mathrm{H})}_{a_1 a_2,\, b_1 b_2} = \mathbb{E}\{[\hat{\boldsymbol{R}}_X - \boldsymbol{R}_x]_{a_1 a_2} [\hat{\boldsymbol{R}}_X - \boldsymbol{R}_x]^*_{b_1 b_2}\}, \quad (11a)$$

and

$$\mathcal{C}^{(\mathrm{T})}_{a_1 a_2,\, b_1 b_2} = \mathbb{E}\{[\hat{\boldsymbol{R}}_x - \boldsymbol{R}_x]_{a_1 a_2} [\hat{\boldsymbol{R}}_X - \boldsymbol{R}_X]_{b_1 b_2}\}, \quad (11b)$$

for each $a_1, a_2, b_1, b_2 \in \{1, \ldots, N\}$. Therefore, to approximate (1) and (2) we use the estimates $\hat{\theta}_i$ and $\tilde{\boldsymbol{R}}_X$, respectively. To estimate (3), we use Wick's factorization, which implies that when $\hat{\boldsymbol{R}}_X = \frac{1}{T} \boldsymbol{X}\boldsymbol{X}^{\mathsf{H}}$ and $\boldsymbol{x}(t)$ is Gaussian, then $\mathcal{C}^{(\mathrm{H})}_{a_1 a_2,\, b_1 b_2} = \frac{1}{T} [\boldsymbol{R}_X]_{a_1 b_1} [\boldsymbol{R}_X]_{b_2 a_2}$ and $\mathcal{C}^{(\mathrm{T})}_{a_1 a_2,\, b_1 b_2} = \frac{1}{T} [\boldsymbol{R}_X]_{a_1 b_2} [\boldsymbol{R}_X]_{b_1 a_2}$. As in (2), we thus approximate (11), i.e., item (3), as

$$\hat{\mathcal{C}}^{(\mathrm{H})}_{a_1 a_2,\, b_1 b_2} = \frac{1}{T} [\tilde{\boldsymbol{R}}_X]_{a_1 b_1} [\tilde{\boldsymbol{R}}_X]_{b_2 a_2}, \quad (12a)$$

$$\hat{\mathcal{C}}^{(\mathrm{T})}_{a_1 a_2,\, b_1 b_2} = \frac{1}{T} [\tilde{\boldsymbol{R}}_X]_{a_1 b_2} [\tilde{\boldsymbol{R}}_X]_{b_1 a_2}. \quad (12b)$$

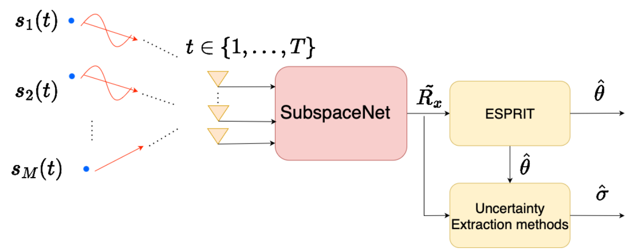

Fig. 1: SubspaceNet with uncertainty extraction

Finally, while Theorem 1 characterizes the variance of the estimator, we use its approximation to estimate $\boldsymbol{\sigma}$, assuming that SubspaceNet provides unbiased estimates of the DoAs. The resulting procedure is summarized as Algorithm 1, and illustrated in Fig. 1.

Algorithm 1: SubspaceNet with uncertainty extract

Input: Trained $\boldsymbol{\psi}$; Signal $\boldsymbol{X} = [\boldsymbol{x}(1), \ldots \boldsymbol{x}(T)]$

Compute surrogate covariance $\tilde{\boldsymbol{R}}_X = \mathrm{DNN}(\boldsymbol{X}; \boldsymbol{\psi})$;

Estimate DoAs via $\hat{\boldsymbol{\theta}} = \mathrm{ESPRIT}(\tilde{\boldsymbol{R}}_X)$;

Estimate $\hat{\mathcal{C}}^{(\mathrm{H})}_{a_1 a_2,\, b_1 b_2}$ and $\hat{\mathcal{C}}^{(\mathrm{T})}_{a_1 a_2,\, b_1 b_2}$ via (12);

for $i = 1, \ldots, M$ **do**

  Approximate $\mathrm{var}(\hat{\theta}_i)$ via (10);

  Set $[\hat{\boldsymbol{\sigma}}]_i = \sqrt{\mathrm{var}(\hat{\theta}_i)}$;

return $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\sigma}}$

C. Discussion

Algorithm 1 is particularly suitable for the problem formulated in Section II under challenges *C1-C3*. While our uncertainty measure relies on approximations of derivations that rigorously hold only in specific scenarios (e.g., Gaussian signals for Wick's theorem) and assumes unbiased estimates, the numerical study in Section IV demonstrates that these approximations still lead to faithful uncertainty quantification under coherent sources, miscalibration, and limited snapshots. In contrast to alternative approaches that require architectural modifications such as training additional outputs or adopting Bayesian networks [31], our method is firmly rooted in established mathematical foundations and can be seamlessly applied with a trained SubspaceNet.

Beyond its empirical performance, the ability to extend deep learning-based DoA estimators with principled uncertainty extraction highlights a broader benefit of model-based deep learning over black-box DNNs. By retaining the interpretable subspace structure, our design naturally supports confidence quantification, which is expected to play a central role in downstream processing tasks. In particular, applications involving distributed arrays or tracking of moving targets can substantially benefit from the additional uncertainty information, enabling reliable sensor fusion.

IV. Numerical Study

We next numerically evaluate Algorithm 1. We consider a ULA with $N = 8$ elements, and (unless stated otherwise) generate $T = 100$ snapshots from $M = 2$ Gaussian sources with DoAs drawn uniformly in $[-\frac{\pi}{2}, \frac{\pi}{2}]$. STD values are averaged over

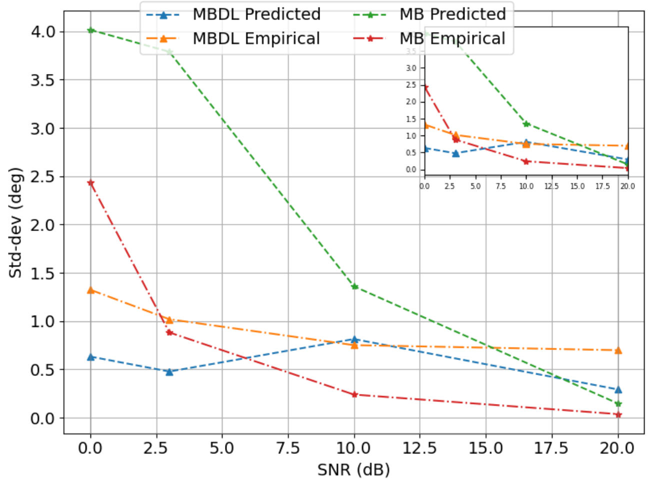

Fig. 2: STD vs SNR, $M = 2$ non-coherent sources, $T = 500$

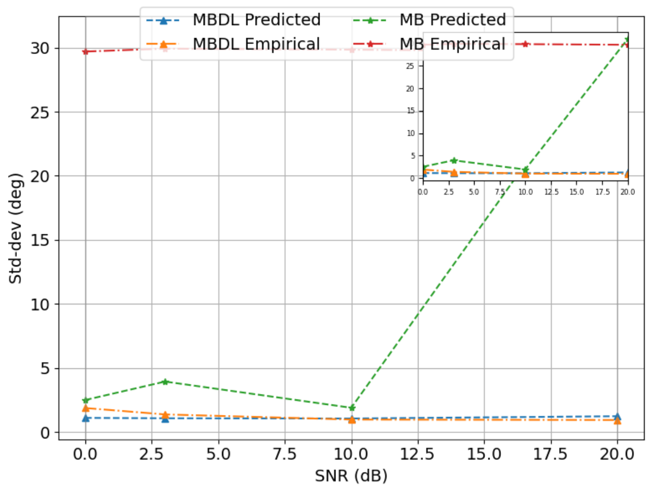

Fig. 3: STD vs SNR, $M = 2$ coherent sources

5000 trials[1]. We implement both model-based ESPRIT (termed *MB*) and SubspaceNet (termed *MBDL*), and for each estimator we compare the predicted $\hat{\boldsymbol{\sigma}}$ obtained via Theorem 1 (for *MB*) and Algorithm 1 (for *MBDL*) to the empirical STD values.

We first consider a simple setting where classic subspace methods are suitable, with $M = 2$ non-coherent sources and calibrated array. The results, reported in Fig. 2, show that when *C1*-*C3* are not encountered, both the *MB* and *MBDL* algorithms accurately estimate the DoAs and predict their error. However, when the sources are coherent, i.e., under *C1*, we observe in Fig. 3 that the *MB* algorithm fails to estimate the DoA and predict its error. However, the *MBDL* Algorithm 1 consistently achieves accurate DoA estimation and faithfully predicts its error,

A similar trend is observed with miscalibration under *C2*. Here, we fix the SNR to 3 dB, and evaluate the algorithms under a deviation of η from half-wavelength element spacing. The results, reported in Fig. 4 show that DNN augmentation does not only increase robustness to miscalibrations (as also noted in [22]), but

[1]The full source code is available online at https://github.com/RazZohar/Uncertainty_SubspaceNet.git.

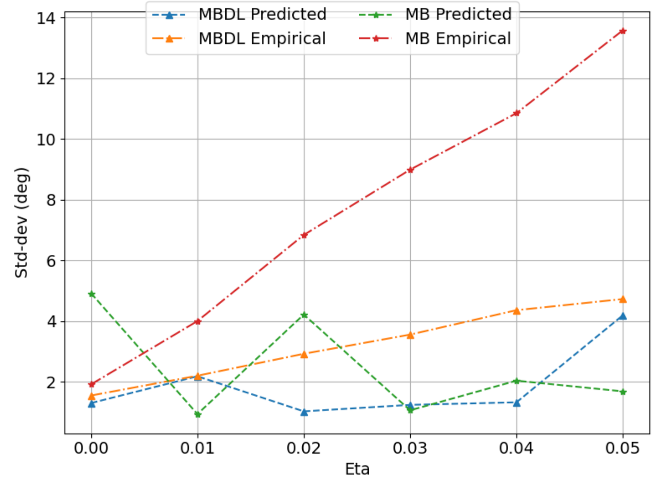

Fig. 4: STD vs calibration offset, $M = 2$ non-coherent sources

that its error can be reliably predicted using Algorithm 1, while the *MB* uncertainty quantification fails to capture its large errors.

Finally, we note that our derivation characterizes the variance of the estimator, which reflects the estimation error when the estimator is unbiased. To show that the characterization is indeed faithful and the bias contribution is relatively minor, we report in Table I the empirical bias, std, and root mean squared periodic error (RMSPE), for all considered settings. There we observe that while the contribution of empirical bias is an order of magnitude lesser than the standard deviation, indicating on the usefulness of our characterization in providing faithful uncertainty metrics.

Case	RMSPE [deg]	Bias [deg]	StdDev [deg]
Non-Coherent $M = 2$	0.538	0.096	0.503
Coherent $M = 2$	0.725	0.097	1.291
Miscalibrated $\eta = 0.02$	1.833	0.533	1.217

TABLE I: MBDL bias and predicted std. SNR $= 3\,\mathrm{dB}$

V. Conclusion

We introduced an uncertainty extraction method for SubspaceNet-based DoA estimation, which leverages its preserved subspace structure to provide uncertainty quantification without architectural modifications. By adapting asymptotic ESPRIT error analysis and identifying suitable approximations based on features provided by SubspaceNet, our method yields faithful uncertainty estimates alongside accurate DoA recovery. Numerical results confirm that the proposed approach enables accurate estimation of both the DoA and the error level in various challenging settings.

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